

● EASY

● ALGEBRA

Linear functions

05

10 questions · ~15 min

Q1

ALGEBRA

Hana deposited a fixed amount into her bank account each month. The function $f(t) = 100 + 25t$ gives the amount, in dollars, in Hana's bank account after t monthly deposits. What is the best interpretation of 25 in this context?

- A. With each monthly deposit, the amount in Hana's bank account increased by \$ 25 .
- B. Before Hana made any monthly deposits, the amount in her bank account was \$ 25.
- C. After 1 monthly deposit, the amount in Hana's bank account was \$ 25.
- D. Hana made a total of 25 monthly deposits.

Q2

GRID-IN ALGEBRA

$$fx = x + \frac{8}{11}$$

The function f is defined by the given equation. What is the value of fx when $x = \frac{3}{11}$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

GRID-IN ALGEBRA

The function f is defined by $fx = \frac{7}{10}x + 55$. What is the value of $f20$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The function h is defined by $hx = x + 200$. What is the value of $h50$?

- A. 200
- B. 250
- C. 10,000
- D. 50,200

The front of a roller-coaster car is at the bottom of a hill and is 15 feet above the ground. If the front of the roller-coaster car rises at a constant rate of 8 feet per second, which of the following equations gives the height h , in feet, of the front of the roller-coaster car s seconds after it starts up the hill?

A. $h = 8s + 15$

B. $h = 15s + \frac{335}{8}$

C. $h = 8s + \frac{335}{15}$

D. $h = 15s + 8$

$$P(t) = 250 + 10t$$

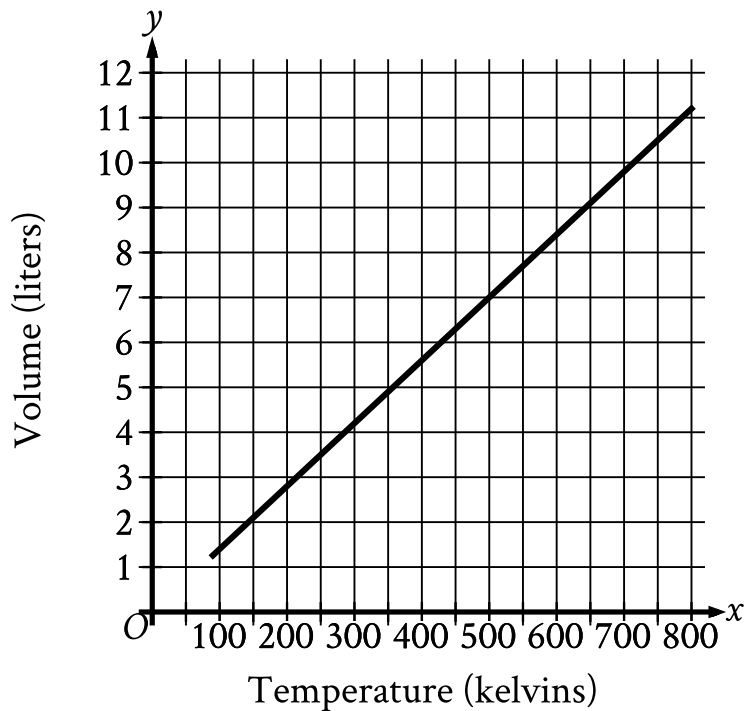
The population of snow leopards in a certain area can be modeled by the function P defined above, where $P(t)$ is the population t years after 1990. Of the following, which is the best interpretation of the equation $P(30) = 550$?

- A. The snow leopard population in this area is predicted to be 30 in the year 2020.
- B. The snow leopard population in this area is predicted to be 30 in the year 2030.
- C. The snow leopard population in this area is predicted to be 550 in the year 2020.
- D. The snow leopard population in this area is predicted to be 550 in the year 2030.

The total cost $f(x)$, in dollars, to lease a car for 36 months from a particular car dealership is given by $f(x) = 36x + 1,000$, where x is the monthly payment, in dollars. What is the total cost to lease a car when the monthly payment is \$ 400?

- A. \$ 13,400
- B. \$ 13,000
- C. \$ 15,400
- D. \$ 37,400

Hydrogen is placed inside a container and kept at a constant pressure. The graph shows the estimated volume y , in liters, of the hydrogen when its temperature is x kelvins.



- The line slants gradually up from left to right.
- The line begins at the approximate point (90 comma 1.3).
- The line passes through the following approximate points:
 - (90 comma 1.3)
 - (500 comma 7)

What is the estimated volume, in liters, of the hydrogen when its temperature is 500 kelvins?

A. 0

B. $\frac{7}{500}$

C. 7

D. $\frac{500}{7}$

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 39 and passes through the point $(0, 0)$. Which equation defines f ?

A. $f(x) = -39x$

B. $f(x) = \frac{1}{39}x$

C. $f(x) = x - 39$

D. $f(x) = 39x$

Area (square feet)	Water (gallons)
2,520	4,536
3,780	6,804
5,040	9,072

The buildings of a shopping center are designed to allow water to drain from the roof into gutters on the sides of the buildings. The table shows the relationship between the area x , in square feet, of a roof and the amount of water fx , in gallons, drained from the roof into the gutters over a certain period of time. Which equation could define f ?

- A. $fx = 0.6x$
- B. $fx = 1.8x$
- C. $fx = 2,268x$
- D. $fx = 4,536x$